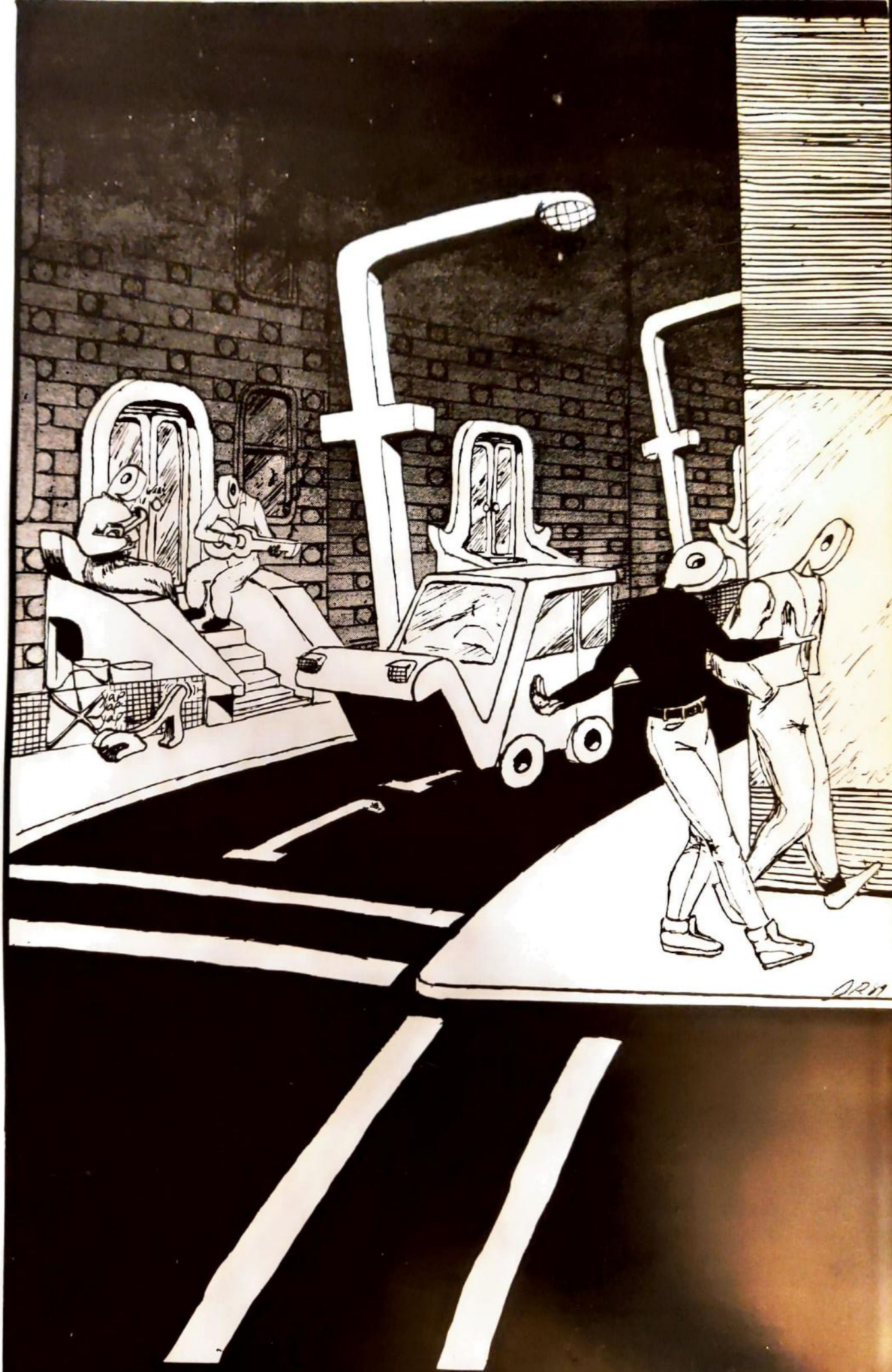


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Talisman Squares

By Jeff Korn

Magic and antimagic squares have properties based on the sums of their rows, columns, and diagonals. Sidney Kravitz, a mathematician from Dover, New Jersey, ignores such considerations and searched for and studied other properties of number arrays. He devised the talisman square, which is defined as an $n \times n$ array of the integers from 1 to n^2 , so that the difference between any one integer and its neighbor is greater than some given constant. A neighboring integer is defined as one immediately adjacent to a given number either vertically, horizontally, or diagonally. The following is a talisman square in which the difference between any integer and its neighbor is greater than one:

1	5	3	7
9	11	13	15
2	4	6	8
10	12	14	16

Figure 1

Note that 13 has 8 neighbors (5,3,7,15,8,4,6,11), and the difference between 13 and its neighbors is always at least 1. Figure 2 shows a talisman square in which the differences are greater than 2.

5	15	9	12
10	1	6	3
13	16	11	14
2	8	4	7

Figure 2

The study of these squares are new in the field of mathematics, and in fact, no rules for construction are known nor are there any mathematical theories that can help in deciding the maximum possible difference between integers and their neighbors. In a 3x3 square containing the integers from 1 to 9, it is impossible to have a difference greater than 1 between all neighboring integers. The integer in the center is surrounded by the other eight and, necessarily, 1 or 2 of these must be numerically consecutive to it.

Several other talisman squares of higher order were found by Kravitz. Figure 3 shows a 5x5 square with differences greater than 4.

15	1	12	4	9
20	7	22	18	24
16	2	13	5	10
21	8	23	19	25
17	3	14	6	11

Figure 3

Talisman squares are only one aspect of this type of problem. Talisman hexagons, such as that shown in figure 4, where the differences are greater than 3 can also be devised:

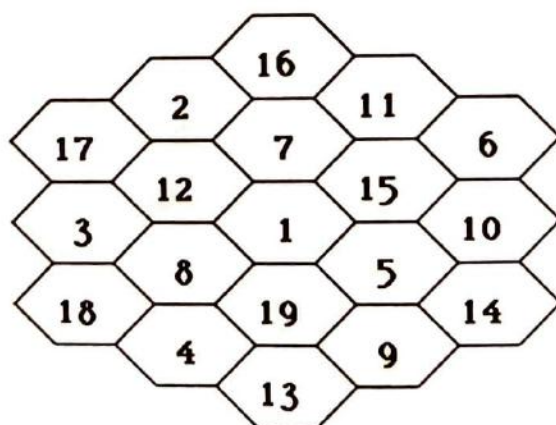


Fig. 4

The problem can be made more complicated if there are more than 8 possible neighbors. For example, one might try to insert the integers from 1 to 36 in a grid of triangles so that differences greater than one exist between any integer and its 12 neighbors.

Basic Solutions in Two-Person Game Theory

by Tony Wong

Game theory is a branch of mathematics that deals with conflicts in terms of their most basic strategic aspects. In a 2 x 2 non-cooperative game (the simplest type), each of two players has two strategies. We can represent this game as follows:

		B	
		I	II
A	I	a	b
	II	c	d

Fig. 1.

where each small letter represents the gain or "payoff" for player A when A and B use the corresponding strategies. A negative payoff means that player B gains.

Three methods of solving such a game will be shown, but these are only needed when there is no obvious solution, or "saddle-point." In the example below, player A will always choose strategy I because he always gains more with it. Consequently, player B will always choose his strategy I to minimize his losses, and there is a saddle-point.

		B	
		I	II
A	I	3	6
	II	2	5

Fig. 2.

When there is no saddle-point, mixed strategies exist. The first method for dealing with this situation is an arithmetic one. Let us use this example:

		B	
		I	II
A	I	a=1	b=5
	II	c=7	d=2

Fig. 3.

A's best strategy mix of I and II is $|c - d| : |a - b|$ or $5 : 4$. Converted to fractions, this mix (which we call X) is $\frac{5}{9} : \frac{4}{9}$. B's optimal ratio is $|b - d| : |a - c|$ or $3 : 6 = 1 : 2$. thus $Y = \frac{1}{3} : \frac{2}{3}$. To find v, the value of the game (the expected payoff to A), we multiply X by a and c and add: $(\frac{5}{9})(1) + (\frac{4}{9})(7) = \frac{33}{9} = \frac{11}{3}$. To check, we can multiply X by b and d and add, and we should get the same answer.

The three formulas for X, Y and v can be generalized as:

$$X = \frac{|c - d|}{|a - b| + |c - d|} : \frac{|a - b|}{|a - b| + |c - d|} = \frac{d - c}{a - b + d - c} : \frac{a - b}{a - b + d - c}$$

$$Y = \frac{|b - d|}{|b - d| + |a - c|} : \frac{|a - c|}{|b - d| + |a - c|} = \frac{b - d}{b - d + c - a} : \frac{c - a}{b - d + c - a}$$

$$v = \frac{a|c - d| + c|a - b|}{|a - b| + |c - d|} = \frac{a(d - c) + c(a - b)}{a - b + d - c} = \frac{ad - bc}{a - b + d - c}$$

In removing the absolute values, we assumed that $a - b$ and $d - c$ have different signs, as do $b - d$ and $a - c$. (If these conditions were not fulfilled, there would be a saddle-point).

A second method takes the matrix in Figure 1 and connects the points (a, c) and (b, d) on the Cartesian plane. Using the example in Figure 3, the result is this:

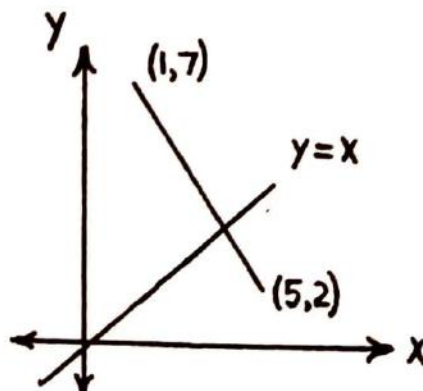


Fig. 4

The line $y = x$ has been drawn to highlight the fact that the segment must intersect that line and have negative slope in order for there to be mixed strategies. When these conditions are satisfied, the value is either coordinate of the point at which the segment intersects $y = x$.

(continued on page 32)

On Which Day of the Week Were You Born?

1. Write the last two digits of the year you were born. Call this number A.
2. Divide that number (A) by 4 and drop the remainder. This answer is B.
3. Find the month in which you were born in the Table of Months below and call that C.
4. Write the date of the month on which you were born. This is D.
5. Now add A, B, C, D.
6. Divide this sum by 7. The remainder from that division should be anywhere from 0 to 6. Find the remainder on the Table of Days. That is the day of the week you were born.

YOU CAN NOT USE THIS METHOD ON DATES BEFORE 1900!

Table of Months

January.....	1	(0 in a leap year)
February.....	4	(3 in a leap year)
March.....	4	
April.....	0	
May.....	2	
June.....	5	
July.....	0	
August.....	3	
September.....	6	
October.....	1	
November.....	4	
December.....	6	

Table of Days

Sunday.....	1
Monday.....	2
Tuesday.....	3
Wednesday..	4
Thursday.....	5
Friday.....	6
Saturday.....	0

A Demonstration of the Equalizing Property of Circles

By David Nowakowski and Ruby Breydo

PROBLEM: To prove that if the centers of the excircles of any triangle are connected to form a new triangle, of that triangle and the centers of the excircles of that triangle are connected to form another triangle, and so forth, the resulting triangles tend to an equilateral triangle.

DEFINITIONS: Given the three (non-concurrent) lines which are the extensions of the sides of a triangle, there are four circles tangent to all three lines:

The *incircle*, which lies in the interior of the triangle.

The *excircles*, the three remaining circles lying outside of the triangle.

The *incenter* and *excenters* are the centers of the incircle and the excircles, respectively.

THEOREM: An excircle lies on the bisector of the angle of the triangle furthest from it, and on the bisectors of the exterior angles closest to it.

PROOF: Draw radii to the points of tangency, (A'R, A'S, A'T). By hypotenuse-leg, $\triangle A'CS = \triangle A'CR$, $\triangle A'BT = \triangle A'BR$, $\triangle A'AS = \triangle A'AT$. Therefore, $\angle A'CS = \angle A'CR$, $\angle A'BT = \angle A'BR$, $\angle A'AS = \angle A'AT$. Repeat the same steps for all excircles.

Using the theorem that the bisectors of two supplementary angles sharing a common vertex and side are perpendicular, we get:

$$m\angle RCA' = 90^\circ - c \text{ and } m\angle RBA' = 90^\circ - b$$

Knowing that the sum of the angles in a triangle is 180° :

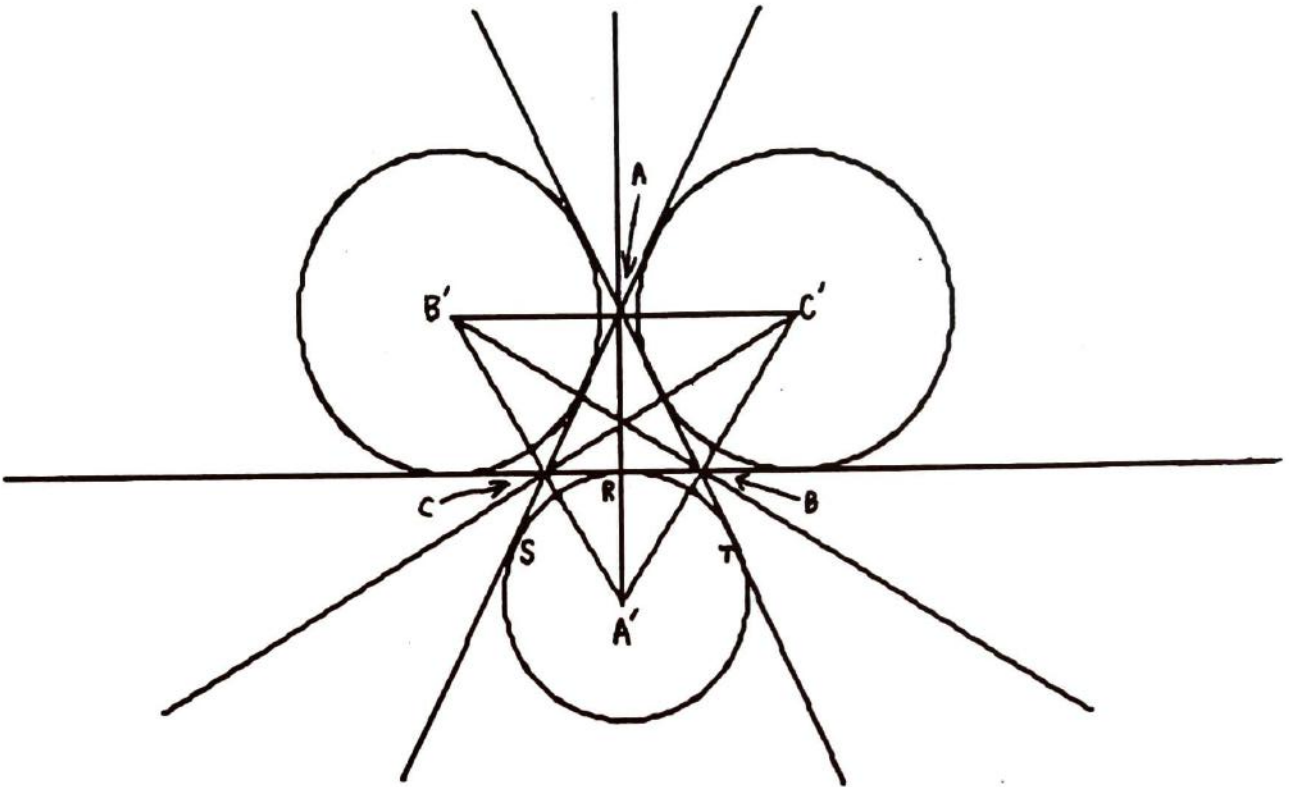


Illustration to The Equalizing Property of Circles.

A Fibonacci Identity

by Amy Zarrin

Have you looked at a tree lately? Or a pineapple? Or a sunflower? If you have, then you were also looking at the Fibonacci sequence. What, you might ask, is that? well, the Fibonacci sequence is a series of numbers that are generated by adding the previous two. the first two terms are 1 and 1, and following terms are 2 (1+1), 3 (1+2), 5 (2+3), 8 (3+5), etc. The n th term is denoted by F_n .

A similar sequence, called the Lucas sequence is generated in the same way, but the second number in this sequence is 3, not 1, and the n th term is denoted by L_n .

Both of these sequences can also be generated by using a formula involving the golden ratio (in fact, the golden spiral is generated using the Fibonacci sequence). First, we let α and β be the roots of the equation x^2-x-1 . These two roots are $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Then, we can use the following formulas, called the Binet forms of the terms:

$$\text{Nth Fibonacci term} = \frac{(\alpha - \beta)^n}{\sqrt{5}}$$

$$\text{Nth Lucas term} = (\alpha + \beta)^n$$

In fact, it is this form that is most often used to prove identities involving this sequence, rather than induction.

And now, on to the identity. In mathematical form, the formula says:

$$(-1)^{m+1}[(F_{km})(L_m) - F_{(k+1)m}] = F_{(k-1)m}$$

Don't let this formidable equation scare you. In English, it makes a lot more sense. Basically what it says is that you can generate every m th term of the Lucas sequence and the previous multiple of the Fibonacci sequence. An example will help clarify this. To generate the tenth term in the Fibonacci sequence, we multiply the fifth term

of both the Fibonacci and the Lucas sequences. To get the fifteenth term, we multiply the fifth Lucas number by the tenth Fibonacci number and add the fifth Fibonacci number. (See chart for more examples and clarification).

Proving this identity involves changing the F_n and L_n terms to the corresponding Binet forms and simplifying. You can find your own identities just by fooling around with the numbers until a pattern emerges.

The Fibonacci Five Pattern

The number added to the subscript (m) = 5

The multiplication number (L_m) = 11

K	K_m	F_{K_m}	$(F_{K_m})(L_m)$	$F_{(K+1)m}$	Difference = $F_{(K-1)m}$
1	5	5	55	55	$0 = F_0$
2	10	55	605	610	$5 = F_5$
3	15	610	6,710	6,765	$55 = F_{10}$
4	20	6,765	74,415	75,025	$610 = F_{15}$
5	25	75,025	825,275	832,040	$6,765 = F_{20}$
6	30	832,040	9,152,440	9,227,465	$75,025 = F_{25}$
7	35	9,227,465	101,502,115	102,334,155	$832,040 = F_{30}$
8	40	102,334,155	1,125,675,705	1,134,903,170	$9,227,465 = F_{35}$
9	45	1,134,903,170	12,483,934,870	12,586,269,025	$102,334,155 = F_{40}$

Math Team: Behind the Scenes

This year's precedent-setting, record-breaking math team will surely go down in history. After years of being paid by Bronx Science to lose, the junior and senior teams finally pulled ahead to victory. Sanjoy Dasgupta managed to achieve the lowest AHSME score in history, and Jun Shimada's alarm clock failed to go off on one hundred and nineteen consecutive days, thus earning him a permanent place in the Guinness Book of World Records.

But many of us mere mortals wonder what goes on in the private lives of the math jocks we see on television. This is why *Math Survey* sent investigative reporter Hillary Anger to uncover the true nature of the math team. Unfortunately, she is on vacation with one of the captains (we're not quite sure which one), so we didn't have a chance to interview her. We then tried to find the coach, Mr. Geller, but we were shocked to find that he was not in any of the local Italian restaurants, or in any of the local pool halls. So, early one morning at 8:00, while it was still dark outside, we visited room 517 ourselves.

It seems that there is a movement to change the official language of the math team to Russian. Michael Hutchings absent-mindedly began lecturing in Russian, and it was a full ten minutes before he noticed that hardly anyone knew what he was talking about. This did not strike him as particularly unusual.

For the first time, there are two women on the senior A team. "It's kind of a pain," said team member Sanjoy Dasgupta, "because they have to have their own separate bathroom on the Lear jet." As for the other team members, Jun Shimada and Jacqueline Huang were unavailable for comment, while Elizabeth Basch was in jail after mugging someone for his New York Times crossword puzzle. Michael Hutchings was in the hospital in critical condition after dividing by zero on a senior meet.

One wonders if next year's math team will ever be able to distinguish itself after this year's. Never to fear. Juniors Richard Sun and Alfred Lin both hope to win the U.S. Open Eraser Throwing Championship. Sophomore Ruby Breydo intends to enter the International Mathematical Olympiad on both the American and Soviet teams simultaneously.

TERRIFY YOUR TEACHERS!

(How to do natural logarithms in your head)

By Michael Hutchings

Many of the astonishing feats of calculating prodigies are made possible by subtle mathematical tricks. In this article, we will use basic calculus to present a scheme for approximating the natural logarithm of a four-digit number with very high accuracy.

The first fact we will use is that

$$\ln(ab) = \ln(a) + \ln(b).$$

This means that we can find the logarithm of a number if we know the logs of its factors. For example,

$$\ln(6000) = 4\ln(2) + \ln(3) + 3\ln(5).$$

If we memorize the logs of 2, 3, 5, 7, and 10, we can find the logs of quite a few four-digit numbers by simple addition.

Now say we want to find $\ln(a)$. We can find the log of some nearby number b using the above technique. Then, we know that

$$\ln(a) = \ln(b) + m\Delta x,$$

where $\Delta x = (a-b)$ and m is the slope of the secant between the points $(a, \ln(a))$ and $(b, \ln(b))$. The Mean Value Theorem tells us that there is some point c , $a < c < b$, such that $f'(c) = m$. Since the third derivative of $\ln(x)$, $2/x^3$, is very small, we know that the slope of the curve is changing at a nearly constant rate. This means that c is very close to $(a+b)/2$, and m is very close to $f'((a+b)/2)$. Now we recall that the derivative of $\ln(x)$ is simply $1/x$. (This is why natural logarithms turn out to be considerably easier than base ten logarithms.)

From this, we get that m is approximately $2/(a+b)$, and

$$\ln(a) \approx \ln(b) + \frac{2\Delta x}{(a+b)}$$

This approximation is very accurate; for four-digit numbers, the accuracy is usually at least 10^{-8} . If we interpolate from two sides of a and average the results, we get even greater accuracy.

If we are doing this mentally, though, we may not be able to handle that many digits, in which case we can use fewer digits. For example, say we want to approximate $\ln(1989)$ to five significant digits. We know that $\ln(2000) = \ln(2) + 3\ln(10)$. Since we've memorized the logs of 2 and 10,

$$\ln(2000) \approx .69315 + 6.90775 = 7.6009.$$

Now, we use our interpolation to find the correction for 1989.

$$\frac{2\Delta x}{(a+b)} = \frac{-22}{3989} \approx \frac{-55}{10000} = -.0055.$$

Therefore,

$$\ln(1989) \approx 7.6009 - .0055 = 7.5954.$$

The actual answer is 7.59539. Close enough!

The scheme shown here is an improvement on the technique of approximation by increments, which is taught in first-term calculus. In general, with approximation by increments,

$$f(x+\Delta x) \approx f(x) + f'(x)\Delta x.$$

With our technique, we use:

$$f(x+\Delta x) \approx f(x) + f'(x+(\Delta x)/2)\Delta x.$$

Using Continued Fractions To Solve Quadratic Equations

By Edward Valentini

An expression of the form:

$$a_1 + \frac{b_1}{a_2 + \frac{b_2}{a_3 + \frac{b_3}{a_4 + \dots}}}$$

is called a *continued fraction*, where $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ may be any real or complex numbers. The number of terms may be finite or infinite.

Some infinite continued fractions are easy to compute. For example, consider the continued fraction:

$$x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

By examining the fraction, we see that:

$$\begin{aligned} x &= 1 + \frac{1}{x}; \\ x^2 &= x + 1; \\ x &= \frac{1 \pm \sqrt{5}}{2} \end{aligned}$$

Of course, the fraction can only have one value. By observing that x is positive, we get $x = \frac{1 + \sqrt{5}}{2}$, which is the golden ratio.

We can perform this procedure in reverse in order to approximate solutions to quadratic equations. For example, say we are given the equation:

$$x^2 - 4x - 1 = 0$$

$$x = 4 + \frac{1}{x}$$

$$x = 4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \dots}}}$$

$$x \approx 4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4}}}} = 4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{17}}}} = 4 + \frac{1}{4 + \frac{17}{72}} = 4 + \frac{72}{305} \approx 4.2360.$$

This is within one ten-thousandth of the actual value.

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Regents Refresher

Due to the New York State Board of Regents' introduction of a new high school mathematics curriculum, it is more difficult for us to provide overview of the material covered than in previous years. Readers who are in the classic courses—algebra, geometry, and trigonometry, designated M1/2, M3/4, and M51/52—should have no difficulty in recognizing the material required for their exams. For the benefit those in sequential math—MQ1/2, MQ3/4, and MQ5/6—we have prefixed the appropriate sections with roman numerals, thus indicating the year in which the material is taught.

This section does not pretend to be a complete review. It includes only the most important theorems and formulae.

(II,III) Equations of Plane Curves:

Line: $y=mx+b$ where m = slope and b = y-intercept.

$y-y_1 = m(x-x_1)$ where (x_1, y_1) is a point on the line and m is the slope.

Slopes of parallel lines are equal.

The product of the slopes of two perpendicular lines is -1.

Circle: $(x-h)^2 + (y-k)^2 = r^2$, where (h,k) is the center of the circle with radius r .

Ellipse: $x^2/a^2 + y^2/b^2 = 1$, where a and b are the major and minor axes.

Parabola: $(x-h)^2 = 2p(y-k)$, where (h,k) is the vertex and p is the distance from the vertex to the directrix

or $y = ax^2 + bx + c$, where $x = -b/2a$ is the axis of symmetry.

Hyperbola: $x^2/a^2 - y^2/b^2 = 1$, where a and b are the major and minor axes.

(III) Logarithms: " $\log_b N = E$ " means $b^E = N$.

$$\log_c 1 = 0$$

$$\log_c c = 1$$

$$\log_c c^b = b$$

$$\log_c a^b = b \log_c a$$

$$\log_c ab = \log_c a + \log_c b$$

$$\log_c 1/a = -\log_c a$$

$$\log_c (a/b) = \log_c a - \log_c b$$

$$\log_c \sqrt[b]{a} = (\log_c a)/b$$

$$\log_b a = \log_c a / \log_c b = \log_b c / \log_a c$$

(II) Laws of Exponents:

$$x^m \cdot x^n = x^{m+n} \quad (x^m)^n = x^{mn} \quad x^m / x^n = x^{m-n} \quad x^{-m} = 1/x^m \quad x^0 = 1$$

$$x^{1/m} = \sqrt[m]{x} \quad x^{m/n} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

(II) The Quadratic Formula:

The roots of the equation $ax^2 + bx + c$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

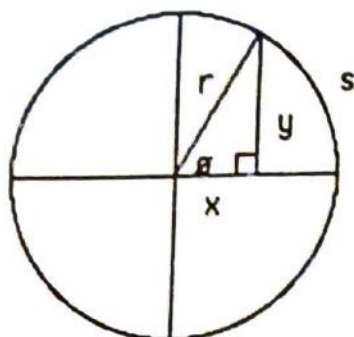
If the discriminant, $b^2 - 4ac$, is positive: the roots are real and distinct.

zero: the roots are real and equal.

negative: the roots are complex conjugates.

(III) Trigonometry:

SOH-CAH-TOA



$$s = r\theta$$

$$r^2 = x^2 + y^2$$

$$\sin\theta = y/r$$

$$\cos\theta = x/r$$

$$\tan\theta = y/x$$

$$\csc\theta = r/y = 1/\sin\theta$$

$$\sec\theta = r/x = 1/\cos\theta$$

$$\cot\theta = x/y = 1/\tan\theta$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\cot^2\theta + 1 = \csc^2\theta$$

$$\sin\theta = \cos(90^\circ - \theta)$$

$$\tan\theta = \cot(90^\circ - \theta)$$

$$\sec\theta = \csc(90^\circ - \theta)$$

Sum/Difference Formulae:

$$\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Double Angle Formulae:

$$\sin(2x) = 2\sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$= 2\cos^2 x - 1 = 1 - 2\sin^2 x$$

$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

(Note: the double angle formulae can be derived from the sum formulae by letting $x=y$.)

Half Angle Formulae:

$$\sin^2(x/2) = \frac{1 - \cos x}{2} \quad \cos^2(x/2) = \frac{1 + \cos x}{2}$$

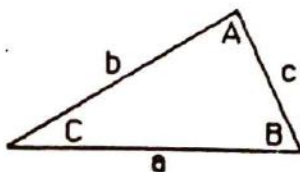
$$\tan^2(x/2) = \frac{1 - \cos x}{1 + \cos x}$$

In the graph of $y = N\sin(rx)$, the amplitude is N , and the period is $2\pi/r$.
360 degrees = 2π radians.

To convert from degrees to radians, multiply by $\pi/180$.

To convert from radians to degrees, multiply by $180/\pi$.

Triangles:



$$\text{Area} = (1/2)ab\sin C$$

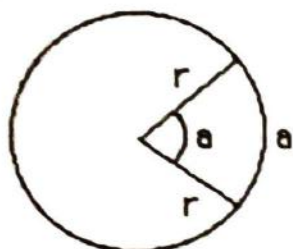
$$= (1/4)\sqrt{(a+b+c)(a+b-c)(b+c-a)(c+a-b)}$$

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \text{diameter of circumcircle}$

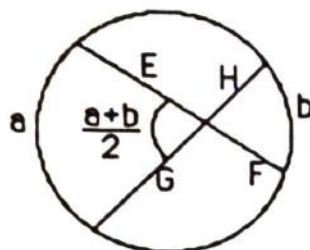
Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$

Law of Tangents: $\frac{\tan((A+B)/2)}{\tan((A-B)/2)} = \frac{a+b}{a-b}$

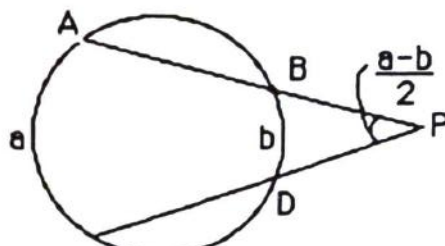
(III) Geometry:



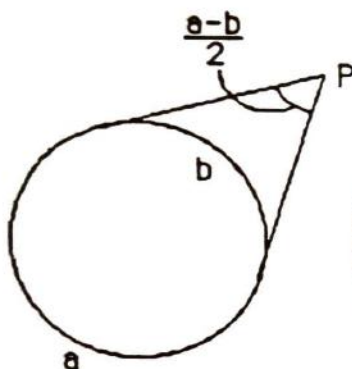
$r = \text{radius}$



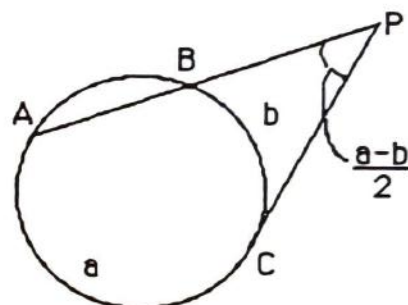
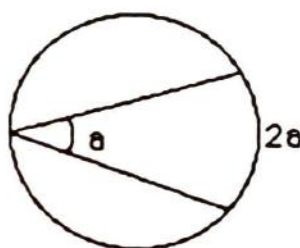
$EF = GH$



$(PA)(PB) = (PC)(PD)$



$PA = PB$



$(PA)(PB) = (PC)^2$

The required proofs for the Geometry Regents may be found in the Dressler text, pp. 733-737. The following are the numbers of the required proofs; 3,5,6,7,8,9,10,11,12,13.

(II) (A=Angle, S=Side, b=base, h=altitude)

Two triangles are congruent if you can prove SSS, SAA, SAS, or ASA.

Two triangles are similar if you can prove AA.

Two right triangles are congruent if you can prove their hypotenuses are congruent and one pair of legs are congruent.(HL)

The sum of the angles in a triangle is 180° .

The base angles of an isosceles triangle are congruent.

Area of triangle $= (\frac{1}{2})bh$.

Area of a trapezoid $= (\frac{1}{2})(b_1 + b_2)h$.

Area of a parallelogram $= bh$.

Area of a regular n-gon $= (\frac{1}{2})ap$ (where a=apothem, p=perimeter).

In a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$.

In a circle, the diameter that is perpendicular to a chord bisects the chord.

The locus of points at distance r from point P is the circle with center P and radius r.

The locus of points equidistant from two points P and Q is the perpendicular bisector of PQ.

The locus of points equidistant from two lines l and m are the bisectors of the angles formed by l and m.

(B=area of base, h=height, r=radius)

Volume of a prism $= Bh$

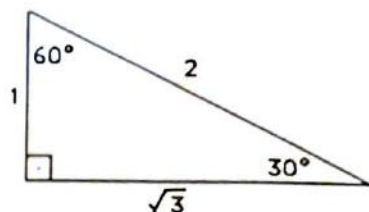
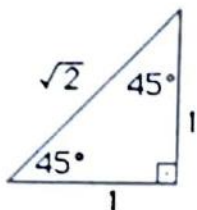
Volume of a cylinder $= \pi r^2 h$

Volume of a pyramid $= (\frac{1}{3}) Bh$

Volume of a cone $= (\frac{1}{3}) \pi r^2 h$

Volume of a sphere $= (\frac{4}{3}) \pi r^3$

The required constructions for the Geometry Regents may be found in the Dressler text, pp. 616-639. The following are the numbers of the required constructions; 5-7, 12, 13, 18-22, 25, 27-29.



(I, III) Probability:

Let A and B represent events in a sample space.

If A and B are independent, then $P(A \cap B) = P(A) \cdot P(B)$.

If A and B are disjoint, then $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$.

If the probabilities are uniformly distributed, event A can occur in $n(A)$ ways, and there are $n(S)$ possible outcomes in S, then $P(A) = n(A)/n(S)$.

The number of permutations of n objects is $n! = n(n-1) \cdots (2)(1)$.

The number of permutations of n objects, k of which are indistinguishable, is $n!/k!$.

The number of permutations of r objects chosen from a set of n objects is

$${}_nP_r = n!/(n-r)! = n(n-1)(n-2) \cdots (n-r+1).$$

The number of combinations of r objects chosen from a set of n objects is

$${}_nC_r = n!/r!(n-r)! = {}_nP_r/r!.$$

(III) The binomial theorem:

$$(a+b)^n = \sum_{r=0}^n {}_nC_r a^r b^{(n-r)} = a^n + {}_nC_1 a^{n-1} b + {}_nC_2 a^{n-2} b^2 + \cdots + {}_nC_{n-1} a^1 b^{n-1} + b^n$$

(I,II) Logic:

Laws of Inference

Symbolic Forms

1. Law of Detachment
(Modus Ponens)

$$[(p \rightarrow q) \wedge p] \rightarrow q$$

2. Law of the Contrapositive

$$(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$$

3. Law of Modus Tollens

$$[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$$

4. Chain Rule (Law of Syllogism)

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

5. Law of Disjunctive Inference

$$[(p \vee q) \wedge \sim p] \rightarrow q$$

$$[(p \vee q) \wedge \sim q] \rightarrow p$$

6. Law of Double Negation

$$\sim(\sim p) \leftrightarrow p$$

7. De Morgan's Laws

$$\sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$$

$$\sim(p \vee q) \leftrightarrow (\sim p \wedge \sim q)$$

The Axiom of Choice

by Alexander Ng

The Axiom of Choice has an interesting place in the history of 20th-century mathematics. It is a set theoretic statement that provoked much controversy when first formulated around 1900. This article will illustrate the usefulness of the Axiom of Choice and describe some startling consequences, such as the Banach - Tarski paradox (explained below).

As you know, mathematicians try to build mathematics from as simple a foundation as possible. The basic goal is to start with a minimal number of axioms and definitions, and then to derive everything else. Set theory and logic form that basic foundation. For example, you can construct the integers by defining:

$0 = \{ \}$ the empty set

$1 = \{ 0 \}$ the set containing zero

$2 = \{ 0, 1 \}$ the set containing zero and one.

etc.

The real numbers, in turn, can be constructed from integers, although this task is non-trivial. In 1872, Robert Dedekind was the first to do this successfully. But I digress.

The Axiom of Choice says the following:

Suppose you have a collection of nonempty sets; then you can pick one element from every set in the collection and put all these "picked" elements in another set.

That's it!

To illustrate, suppose some old lady is standing in the fruit aisle of a supermarket. There are fruit bins along the aisle, all nonempty. The Axiom of Choice says the lady can fill her basket with fruit so that it contains exactly one item from each bin. There are two important questions raised by the Axiom of Choice. First, is

the Axiom of Choice true when the collection of sets contains infinitely many sets? Second, can the Axiom of Choice be derived as a theorem from the other seven accepted set theory axioms?

In 1966, P.J. Cohen proved that the Axiom of Choice cannot be derived from the other axioms. It is not a theorem. But Kurt Godel was able to show that the addition of this axiom to the existing set theory does not produce a contradiction. This property is called *consistency*. The Axiom of Choice is accepted by most mathematicians without qualms, primarily because many useful desirable theorems are consequences.

However, it is possible to construct a mathematical system which denies the Axiom of Choice, but accepts the remaining axioms. Such a mathematical system is analogous to non-Euclidean geometry. Riemannian geometry is one example; it rejects the parallel postulate and instead says:

Through a given line L and a point P off the line L , there does not exist a line passing through P and parallel to L .

Consequences of the Choice Axiom:

1. **Banach - Tarski Paradox:** This "paradox" is actually a theorem. It is one of the strangest and most upsetting consequences of the Axiom of Choice. The main idea is that given a solid sphere of some volume, it is possible to cut it up into six pieces and reassemble the sphere into one with any arbitrary volume. That is like saying you can cut up a Sunkist orange into six pieces and glue it together to form an orange the size of the sun. Clearly, the Banach - Tarski paradox does not hold in real life. This is one reason mathematicians have objections to accepting the truth of the Axiom of Choice.

2. **Discontinuous Additive Functions.** Consider a function $f(x)$ whose domain and range are of the set of real numbers. We say that $f(x)$ is *additive* if for all x, y in R : $f(x + y) = f(x) + f(y)$. For any constant c , the linear function $f(x) = cx$ obviously

has this additive property. In fact these are the *only* continuous functions which are additive.

A stimulating question that Cauchy considered in the early 1800's was whether there are any discontinuous additive functions. The answer is yes. In 1905, G. Hamel used the Axiom of Choice to prove that such functions exist. Although it is not possible to explicitly write down such a function, the graph is very interesting. Let a white piece of paper with x and y axes represent the coordinate plane. If you use a black pencil to draw the graph, then you must shade the entire sheet black to represent the graph. Such a function has a *dense* graph.

There are other important theorems that are logically equivalent to the Axiom of Choice. They are called Zorn's Lemma, Hausdorff Maximality Principle, Well-Ordering Theorem and Tychonoff's Theorem. These other statements are theorems of the Axiom. However, if you assume these other results, you can derive the Axiom of Choice.

As you can see, the Axiom of Choice is both a bizarre and powerful tool with a turbulent history. It shakes the foundation of mathematics while giving beautiful results. You may find research into this Axiom and other problems like the Continuum Hypothesis interesting. This field is called axiomatic set theory.

Weird. All very weird ...

*Best Wishes
from New York University
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of Mathematical Sciences*

The Evil That Teaches

by Fumbeya Marungo

You are entering another dimension. A dimension not only of sight and sound, but of mind...

It's the eve of the first day of school and John Tang, a Stuyvesant senior is going to bed. But this will not be an ordinary night, for John is about to see what every Stuy student fears, in the Twilight Zone...

After watching a little TV, John goes to bed. As he relaxes, he begins to wonder who his calc teacher will be. Suddenly, he is in a lime green room. He quickly realizes that he is at Stuy (swift, ain't he?). The scene is blurry, but John can make out a faint shadow in the background. An ominous voice says, "What is the derivative of $6x^3+4x^2+8x+1$."

"I don't know," meekly responds John.

The voice booms, "You don't know?! You don't know?! You moron! You idiot! You couldn't get into Pace University. What do you think this is? John Dewey High School! The Knicks have a better chance of winning the playoffs than you of graduating, you fool!"

The voice reverberates in John's head, he swoons this way and that. He desperately tries to get a look at the figure's face, but to no avail. Suddenly, he sees a number; it's his calculus grade! The voice is right, he won't even get into Pace! Queens Community might take him, if he's lucky! The face is beginning to clear, John looks to see who his tormentor is. It's... it's...

John wakes up, panting, drenched in sweat. He has been saved from continuing torment by his loving alarm clock. He gets ready for school, still shaken by the dream.

As he enters homeroom, he gets his program. Checking to see who his teachers are, he realizes that his calc teacher is... is... Oh, noooooooooo!!!!!!!!!!!!

It's been said that teachers are benevolent pedagogues, but John Tang knows better. He has learned this valuable lesson in... The Twilight Zone.

Testing Divisibility by Primes

by Michael Hutchings

We may wish to determine whether or not a prime number p is a factor of x . A divisibility test is an algorithm which gives a smaller number x' which is divisible by p if and only if x is. In this article, we will present divisibility tests for some small primes and present a general divisibility test for all other primes.

Theorem 1: A number is divisible by 3 if and only if the sum of its digits is.

Lemma 1: For all positive integers n ,

$$10^n = 9x_n + 1,$$

for some x_n . We will prove this by induction.

Clearly, this is true for $n=1$. Now assume that for some k :

$$10^k = 9x_k + 1.$$

We now know that:

$$\begin{aligned} 10^{k+1} &= 10(9x_k + 1) \\ &= 90x_k + 10 \\ &= 9(10x_k + 1) + 1. \end{aligned}$$

Setting $x_{k+1} = 10x_k + 1$, we get:

$$10^{k+1} = 9x_{k+1} + 1.$$

Proof of Theorem 1: Given any number x ,

$$x = a_0 + 10a_1 + 10^2a_2 + \dots + 10^na_n,$$

where $a_0, a_1, \dots, a_n$ are the decimal digits. By Lemma 1,

$$\begin{aligned}
 x &= a_0 + (9x_1 + 1)a_1 + (9x_2 + 1)a_2 + \dots + (9x_n + 1)a_n \\
 &= (a_0 + a_1 + \dots + a_n) + 9(x_1a_1 + x_2a_2 + \dots + x_na_n).
 \end{aligned}$$

From this, it is clear that x is a multiple of 3 if and only if the sum of its digits is. We can also see that the divisibility rule for 9 is the same.

Theorem 2: A number is divisible by 11 if and only if the sum of the even digits $a_0, a_2, a_4, \dots$ and the sum of the odd digits differ by a multiple of 11.

Proof: We can express a decimal number as a polynomial

$$P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n.$$

For our purposes, $x = 10$. We wish to see if 11 is a factor of $P(x)$, so we will divide $P(x)$ by $(x+1)$. By the Remainder Theorem, $P(-1)$ is the remainder when $P(x)$ is divided by $(x+1)$. For our number to be divisible by 11, this remainder must be either 0 or a multiple of 11. Thus,

$$\begin{aligned}
 P(-1) &= 11k; \\
 a_0 + a_1(-1) + a_2(-1)^2 + a_3(-1)^3 + \dots + a_n(-1)^n &= 11k; \\
 (a_0 + a_2 + a_4 + \dots) - (a_1 + a_3 + a_5 + \dots) &= 11k.
 \end{aligned}$$

Theorem 3: Given any number $x = 1000a + b$ (where b represents the last three digits and a is the rest of the number), x is divisible by 7 if and only if $(a-b)$ is also divisible by 7. The same holds true for 11 and 13.

Proof: Note that:

$$\begin{aligned}
 1000a + b &= 1000(a-b) + 1001b; \\
 1000a + b &= 1000(a-b) + (7)(11)(13)b.
 \end{aligned}$$

$(1000a + b)$ is a multiple of 7 if and only if $1000(a-b)$ is. Since 1000 contains no factors of 7, this means that $(a-b)$ is a multiple of 7. The same argument holds for 11 and 13.

This type of divisibility test can be found for other primes. The reader may verify the following:

7: If $x = 10a + b$, $x' = a - 2b$.

17: If $x = 100a + b$, $x' = 2a - b$. Or, if $x = 1000a + b$, $x' = a - 6b$.

19: If $x = 10000a + b$, $x' = a - 3b$.

23,29: If $x = 1000a + b$, $x' = a - 2b$.

However, this technique does not make a good general divisibility test, because these formulas are not always easy to find. We will now present a general divisibility test which works by dividing from right to left by any number that is not a multiple of 2 or 5. (Most primes fall into this category.) This form of division is much faster than long division, and, if successful, also gives the quotient, unlike the other divisibility tests.

We will first give an example of the algorithm and then prove its validity. To check if 160047299 is a multiple of 17, the work is as follows:

$$\begin{array}{rcl}
 160047299 & & \\
 -119 & = & 7 \cdot 17 \\
 \hline
 16004718 & & \\
 -68 & = & 4 \cdot 17 \\
 \hline
 1600465 & & \\
 -85 & = & 5 \cdot 17
 \end{array}$$

$$\begin{array}{rcl}
160038 & & \\
-68 & = & 4 \cdot 17 \\
\hline
15997 & & \\
-17 & = & 1 \cdot 17 \\
\hline
1598 & & \\
-68 & = & 4 \cdot 17 \\
\hline
153 & = & 9 \cdot 17
\end{array}$$

Since we have reduced our number to a multiple of 17, the original number was a multiple of 17. In addition, we see that the quotient is 9414547.

Algorithm 1 (Right-Left Integer Division): In general, to check if x is divisible by p (p is not a multiple of 2 or 5), we find x' as follows:

1. Find k , $0 \leq k \leq 9$, such that the last digit of pk is the same as the last digit of x . (To find k , one need only consider the last digit of p .)
2. Subtract pk from x .
3. Remove the rightmost 0 from x .

Repeat this process until:

Case 1: Zero is reached. In this case, x is divisible by p , and the k 's, going backward, spell out the quotient.

Case 2: A number known not to be a multiple of p is reached. In this case, x is not a multiple of p .

Theorem 4: Algorithm 1 works.

Proof: First, we must show that it is possible to find k , as in step 1 of the algorithm. Assume that no such k exists. Let y be the last digit of x . If $y = 0$, then $k = 0$, so we will ignore that case. Thus, y is between 1 and 9. Now, let d_k equal the last digit of pk . If there is no k such that $d_k = y$, then, by the Pigeon-Hole Principle, there must exist i and j (without loss of generality, $i > j$) such that $d_i = d_j$. If this is so, then $(pi - pj) = p(i - j)$ is a multiple of 10. $(i - j)$ is clearly between 1 and 8, and is thus not a

multiple of 10. This means that p must contain factors of 2 or 5. But it doesn't, so this is a contradiction, our assumption must be false, and k therefore exists.

Next, we need to show that x' is a multiple of p if and only if x is.

$$x' = \frac{x - kp}{10}, \text{ and hence, } x - kp = 10x'.$$

If x is a multiple of p , then so is $(x - kp)$. Since this is equal to $10x'$, $10x'$ is therefore a multiple of p . p has no common factors with 10, and x' is therefore a multiple of p . Similarly, if x is not a multiple of p , then neither is $(x - kp)$. Thus, $10x'$ is not divisible by p , and as a result, x' is not divisible by p .

Finally, we will show that the k 's spell out the quotient. We will say that

$$\frac{x}{p} = a_0 + 10a_1 + 10^2a_2 + \dots + 10^na_n, \text{ and}$$

$$x = pa_0 + 10pa_1 + 10^2pa_2 + \dots + 10^npa_n,$$

where $a_0, a_1, \dots, a_n$ are the decimal digits of the quotient. Since the last digit of a_0p is the last digit of x , the first value of k is clearly a_0 . Then,

$$\begin{aligned} x' &= \frac{x - kp}{10} = \frac{10pa_1 + 10^2pa_2 + \dots + 10^npa_n}{10} \\ &= pa_1 + 10pa_2 + 10^2pa_3 + \dots + 10^{n-1}pa_n. \end{aligned}$$

This means that the next value of k will be a_1 . Continuing this process, we will get all the digits of the quotient as values of k .

8. Law of Simplification

$$(p \wedge q) \rightarrow p$$

$$(p \wedge q) \rightarrow q$$

9. Law of Conjunction

$$(p) \wedge (q) \rightarrow (p \wedge q)$$

10. Law of Disjunctive Addition

$$p \rightarrow (p \vee q)$$

(I, III) Statistics

Measures of Central Tendency:

The mean, or arithmetic mean, of a set of n numbers is the sum of the numbers divided by n . Therefore, for a sample of scores, $x_1, x_2, x_3, \dots, x_n$:

$$\text{The mean} = \bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

When a set of data is arranged in numerical order, the "middle" score is called the median. If there is an even number of terms, then the median is found by averaging the two "middle" terms.

The mode is the score that appears most often in a set of data. For some sets of data there can be more than one mode (when two modes appear, the data is called bimodal) and, in some cases, no mode whatsoever.

Measures of Dispersion:

The range of a set of data is the difference between the highest and the lowest scores.

The mean absolute deviation of a set of data is the average value of the difference between the data and the mean. In symbols:

$$\text{mean deviation} = (1/n) \sum_{i=1}^n |x_i - \bar{x}|$$

The variance, v , of a set of data is the average of the squares of the deviations from the mean. In symbols:

$$\text{variance} = v = (1/n) \sum_{i=1}^n (x_i - \bar{x})^2$$

The standard deviation, s , of a set of data is equal to the square root of the variance. In symbols:

$$\text{standard deviation} = s = \sqrt{v}$$

$$m\angle C'A'B' = 180 - (m\angle RBA' + m\angle RCA') = b + c$$

Repeating with the other circles:

$$m\angle A'C'B' = a + b \quad \text{and} \quad m\angle A'B'C' = a + c$$

In any non-equilateral triangle, there must be an angle greater than 60° . Thus, call the angles:

$$60^\circ + x, 60^\circ - y, \text{ and } 180 - ((60 + x) + (60 - y)) = 60^\circ + y - x.$$

By the observation made above, the angles of the triangle after connecting the excenters once will be:

$$60^\circ + \frac{y}{2} \qquad 60^\circ - \frac{x}{2} \qquad 60^\circ + \frac{x}{2} - \frac{y}{2}.$$

Connecting the excenters n times, we get:

$$60^\circ \pm \frac{y}{2^n} \qquad 60^\circ \pm \frac{x}{2^n} \qquad 60^\circ \pm \frac{x}{2^n} \pm \frac{y}{2^n}.$$

$$\lim_{n \rightarrow \infty} 60^\circ \pm \frac{y}{2^n} = 60^\circ \quad \lim_{n \rightarrow \infty} 60^\circ \pm \frac{x}{2^n} = 60^\circ \quad 180 - (60 + 60) = 60^\circ.$$

Q.E.D.

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(continued from page 6)

This can be proven as follows:

1. The slope of the segment is $\frac{(d - c)}{(b - a)}$, so its equation is

$$y - c = \frac{(d - c)}{(b - a)} (x - a) \text{ or } y = \frac{(d - c)}{(b - a)} (x - a) + c$$

2. Now, setting $y = x$ and multiplying by $(b - a)$:

$$x(b - a) = (d - c)(x - a) + c(b - a)$$

$$= (d - c)x - (d - c)a + c(b - a)$$

$$x(b - a + c - d) = c(b - a) - a(d - c)$$

$$= cb - ad$$

$$x = \frac{cb - ad}{-a + b - d + c} = \frac{ad - bc}{a - b + d - c}$$

Player B's optimal strategy is given as the ratio into which the segment is divided. (This can be proven using the formula for dividing a segment between two points into a given ratio.) Player A's optimal strategy will be $m : 1 - m$ when the equation of the segment is put in the form $mx + (1 - m)y = k$. A more direct way of finding it in a 2×2 game is plotting (a, b) and (c, d) , connecting them, and finding the ratio into which $y = x$ divides the resulting segment. Can you show how both these methods are consistent?

It turns out that the graphical method shown above can be used for any $2 \times n$ game - B will draw a "strategy polygon" of n vertices. The only condition for a mixed strategy is that the side intersecting $y = x$ furthest to the lower left has negative slope. As

a result, one can always eliminate all but two of the columns, or points. In the example of a 2 x 3 game below, the optimal strategy for B is $\frac{1}{2} : \frac{1}{2}$. A's strategy is then found, taking into account only points P and Q.

$$\begin{array}{c} \text{A} \end{array} \begin{array}{c} \text{B} \\ \left[\begin{array}{ccc} 5 & 2 & 6 \\ 2 & 5 & 4 \end{array} \right] \end{array}$$

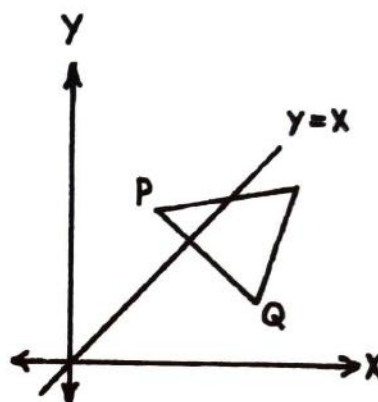


Fig. 5.

The third method for mixed strategies is also a graphical one, but it takes the basic 2 x 2 matrix and transforms it into several equations:

$$\begin{array}{ll} y = (a - c) X + c & y = (a - b) Y + b \\ y = (b - d) X + d & y = (c - d) Y + d \end{array}$$

Here, $X_1 : 1 - X_1$ makes up A's optimal strategy X, and $Y_1 : 1 - Y_1$ make up B's optimal strategy. We graph the first pair in the $X_1 - y$ plane from 0 to 1 (since $0 < X_1 < 1$) and the second pair in the $Y_1 - y$ plane.

Using the same example as in Figure 3, these are the two graphs:

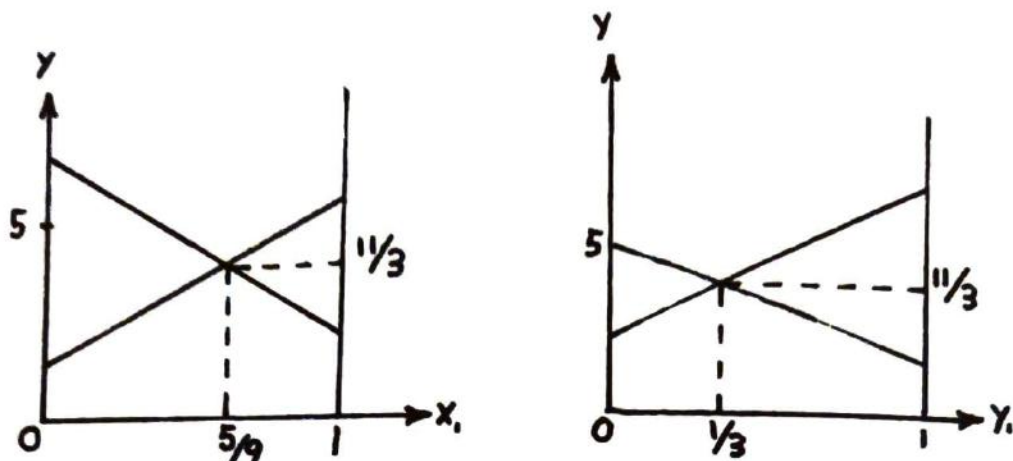


Fig. 6

The value X_1 (or Y_1) at which the two lines intersect is the optimal X_1 (or Y_1). The y-value of the intersection point, in both cases, is the value of the game. One can confirm this method using the basic equations given above - a little manipulation will yield the general formulas.

This method can also be extended to $2 \times n$ games. One draws n segments on the same graph and picks the point that, starting from the bottom and moving up, one will arrive at last without crossing a line segment. Such a point is P in the example below. (If a line segment satisfied this condition, then there would be multiple solutions.)

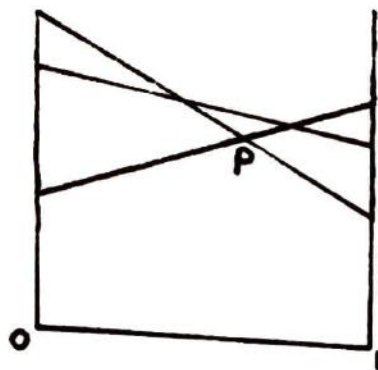
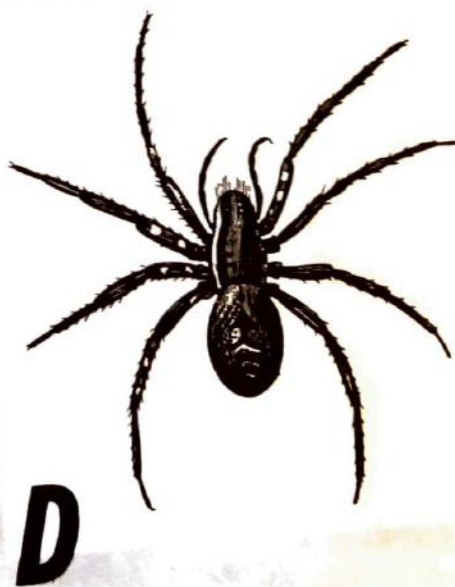


Fig. 7

Would this last method work for 3×3 games, with planes drawn in 3-D space? Perhaps a generalization for any number of dimensions can be created.



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Blind Date Disaster

by Connie (Kang) Lin

Five blind dates were set up by a local dating service. All five couples were to meet at the famous Hard Rock Cafe. As a joke, none of them were told the name of their blind date. Instead, they were told bits of information to help them find their date. The problem is made even more difficult because one of the five guys is a notorious for standing up his dates. The other four guys are a famous and wealthy British new wave singer, a bisexual, a tennis player, and a Stuyvesant student. One and only one member of each couple was given false information. In four of the five couples, the guy has the false information. Cary is a known liar. The only guy that was given the correct information is the Stuyvesant student (of course). A further complication is that all ten people have names that can be used for both sexes. Who is going to get stood up? Here are the bits of information given to each person...

- | | |
|-------|---|
| Alex | 1. Nicki is Robin's date. |
| Brook | 2. Jerry is the singer's date.
3. Alex has false information. |
| Cary | 4. Brook is Hasan's date. |
| Chris | 5. Brook is Terry's date.
6. Cary is the tennis player. |
| Jerry | 7. Chris is not Robin's date. |
| Jesse | 8. Nicki is not the Stuyvesant student.
9. Alex is Chris' date. |
| Nicki | 10. Cary is Hasan's date.
11. Terry is not the bisexual. |
| Hasan | 12. Chris is either the singer, or Chris is a woman.
13. Alex is Jerry's date. |
| Robin | 14. Cary is Jesse's date.
15. Brook is a woman. |
| Terry | 16. Jerry stands women up.
17. Cary tells the truth. |